Grain Definition Document

NYC Ride-Hailing, Weather & EV Charging Access

Course	DL3004 Data Warehousing & BI Lab, Fall 2026, FAST-NU Lahore
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The grain says what one row of a fact table means. We fix the grain first, so every measure and dimension fits it.

1. Grain Statements

Fact Table	Grain Statement	Row Is Unique By	Why This Level
Fact_Trips	**One row represents one completed ride-hailing trip, from pickup to drop-off, in the sampled TLC data.**	trip_id (generated row number)	It is the lowest level in the data. Any summary can be built from it.
Fact_Zone_Hourly_Snapshot	**One row represents one pickup zone in one hour, including hours with zero trips.**	zone_key + date_key + time_key	Weather is measured hourly, so an hourly zone summary makes rain comparisons simple.
Fact_Charger_Coverage	**One row represents one charging station opened in one zone on one date.**	station_id (each station appears once)	It records that coverage exists, with no number to store.

2. Testing the Grain Against Business Questions

Question	Fact Table	Can the Grain Answer It?
Which zones have the most trips each month? (Q1)	Fact_Trips	Yes. Count trip rows by zone and month.
What is the average wait by hour? (Q4)	Fact_Trips	Yes. Use total wait / trip count, not an average of averages.
How much does rain change trips per hour? (Q5)	Fact_Zone_Hourly_Snapshot	Yes. One row per zone-hour already holds trips and rainfall.
Which zones have no charging station? (Q9)	Fact_Charger_Coverage	Yes. Find zones in Dim_Zone with no matching row.
Which individual trip took 90 minutes?	Fact_Zone_Hourly_Snapshot	No. The snapshot has no trip-level rows. Use Fact_Trips.

3. Grain Rules We Follow

- Every fact table has exactly one grain statement.
- We never mix grains in one table. Trip rows and zone-hour rows stay in separate fact tables.
- Every dimension and measure must make sense at its table's grain.
- Totals are stored, and averages are calculated from totals.